

HYDROGEN AND METHANOL ECONOMY COURSE PROJECT PLAN

Techno-Economic Optimization of a Solar-Powered Alkaline Electrolyzer System for Industrial Green Hydrogen Production

1. Problem Identification -

Green hydrogen can be produced by using renewable electricity to split water in an electrolyzer. However, the final cost of hydrogen depends on several factors such as **solar power availability, electrolyzer efficiency and utilization, equipment cost, electricity cost and financing**. The project will study these factors for a alkaline electrolyzer system and identify an economically suitable system configuration for industrial green hydrogen production.

2. Objectives -

The main objective is to develop a simple techno-economic model for a solar-powered alkaline electrolyzer system. The study will cover:

- Estimation of **hydrogen production** and **LCOH (cost of producing 1 kg of hydrogen)**.
- Comparison of different **solar and electrolyzer capacities** to find a suitable configuration.
- Effect of **electricity cost** on hydrogen production cost.
- Effect of **electrolyzer efficiency and utilization**.
- Analysis of **CAPEX and OPEX**.
- Effect of **financing parameters** on the overall project economics.

3. Work Plan -

- Literature Survey:** We will collect published data on alkaline electrolyzers, solar power, hydrogen production, equipment costs, and financing assumptions.
- Model Development:** Then we will develop a calculation model in Excel/Python using the collected data.
- Performance Calculation:** We will calculate solar electricity generation, hydrogen production, water requirement, CAPEX, OPEX, and LCOH.
- Scenario Analysis:** We will test different operating and design conditions to study their effect on hydrogen production cost.
- Optimization:** Then identify the suitable combination of solar and electrolyzer capacity that gives a lower LCOH.
- Machine Learning:** Use the generated results to develop a simple ML model for LCOH prediction based on the main input parameters.
- Validation:** Compare the calculated results with published values to check the reliability of the model.

4. Expected Outcome / Solution -

The project expected solution is to **provide an estimate of the hydrogen production cost for the selected system configuration and identify the main factors controlling its economics**.

5. Proposed Timeline –

Week	Work
Week 1	Literature survey, problem definition and collection of technical/economic data.
Week 2	Develop the calculation model and calculate hydrogen production, CAPEX, OPEX and LCOH.
Week 3	Scenario analysis, and optimization of solar and electrolyzer capacities.
Week 4	ML prediction, validation with published results, final conclusions and report preparation.